

The covar and orig Data-Generating Processes: A Component-Wise

Assessment, with Application to the $c_{bm} \leq 0.35$ Regime

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1. Executive summary

Paper 01 (`analysis/report/01-dgp-mean-moderation-vs-mvn/`) compares three data-generating processes, of which two are covariance-channel processes: `covar` (the package's Architecture B) and `orig` (a vendored copy of the Hendrickson et al. reference implementation). Appendix B.6 of that paper states that `orig` differs from `covar` on three rows simultaneously. This white paper audits that claim against the code that runs, assesses each difference on its merits, and answers a specific practical question: if the biomarker-response correlation c_{bm} is restricted to 0.35 or below, is there any remaining reason to prefer `covar` over `orig`?

There are four differences, not three, and the appendix's account of the most consequential one is inverted. The three documented rows are the within-factor correlation form, the cross-factor off-diagonal, and the interaction vector `b`. A fourth, undocumented difference is that `orig` never assigns the biomarker-response correlation at the first measurement occasion. The appendix's description of `b` corresponds to code that is commented out in the vendored file; the active code behaves in the opposite direction.

Only one of the four changes is required to fix positive definiteness. A crossed decomposition over all 16 combinations of the four changes (Section 5) shows that switching compound symmetry to AR(1) accounts for the entire improvement, raising the worst-cell c_{bm} ceiling from 0.34 to 0.48. Applied alone it beats the full four-change `covar` set, which reaches only 0.45. The cross-factor change lowers the ceiling to 0.16 when applied without AR(1), and neither the `b` change nor the occasion-1 change moves it at all.

The answer to the practical question is yes, on five independent grounds. Restricting c_{bm} to 0.35 does not rescue `orig`. The positive-definite (PD) ceiling under `orig` is 0.30 in five of nine design-by-half-life cells and 0.35 in the remaining four, so a study run at 0.35 is non-PD in a substantial part of its own grid. Where it is PD at 0.35 the margin is thin enough to be fragile to any other parameter change. PD failure is repaired silently, so the realized correlation stops matching the nominal one without any warning. The dropped first occasion is a defect at every c_{bm} , including zero. And the `b` decay rate

differs by a factor of two, so the two processes are not comparable at a matched nominal half-life regardless of correlation strength.

2. Scope, method, and epistemic status

Findings below are labeled **verified** (code was executed and output observed), **inspected** (read in source and confirmed by reading), or **inferred**. Nothing in this document is asserted from memory of the manuscripts.

Sources examined:

- `analysis/scripts/quick-sim/hendrickson-original-comparison/`
(`vendored-hendrickson-generateData.R`, `01-hendrickson-orig-driver.R`, `02-pd-sweep.R`, `README.md`)
- `R/generateData.R` (buildSigma, the covar construction)
- `implementations/original/R/generateData.R`
- `analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`,
Appendix sections B.4 through B.7

Computations run for this document:

- Reconstruction of the `b` vector under both the active and the commented-out code paths, Hybrid path A, at `t_half` in $\{0.5, 1.0\}$.
- A PD sweep over `c_bm` in $[0.10, 0.60]$ by 0.05, crossed with `t_half` in $\{0, 0.5, 1.0\}$ and the three trial designs, recording the minimum eigenvalue of Sigma under each process, minimized across the design's randomization paths.
- Quantification of the distortion introduced by PD repair.

All computations used the package's own `extracted_rp` and `extracted_bp` parameter sets, `rho = 0.7`, `c.cf1t = 0.2`, `c.cfct = 0.1`, `N = 35`, `scalefactor = 2`.

3. The four differences

3.1 Difference 1: within-factor correlation form

Status: **verified**.

`orig` assigns a single constant to every within-factor occasion pair (`vendored-hendrickson-generateData.R`:117-126):

```
ac <- modelparam[[paste("c", c, sep = ".")]]
for (p in 1:(nP-1)) for (p2 in (1+p):nP) {
  correlations[n1, n2] <- ac
  correlations[n2, n1] <- ac
}
```

This is compound symmetry. `covar` uses $\rho^{|w_i - w_j|}$, an AR(1) form on the cumulative week scale.

Arguments for compound symmetry. It is the form in the published reference, so it is the correct target if the goal is reproducing Hendrickson et al. It has one parameter and is exchangeable, which makes it invariant to occasion ordering and therefore robust to irregular visit spacing. It is the standard random-intercept implication, so it is internally consistent with an analysis model that fits only a patient random intercept.

Arguments against. It is clinically implausible for repeated symptom measurement, where adjacent occasions should correlate more strongly than distant ones. More seriously for present purposes, it constrains the feasible parameter space. A constant ρ across n occasions requires $\rho > -1/(n-1)$ on its own, and once the biomarker-response block is added the joint constraint on c_{bm} tightens rapidly with n . Appendix B.7 identifies this as the mechanism behind the narrower PD range, and Section 4 below quantifies it.

Assessment. This difference is the dominant driver of the PD behavior and therefore of the answer to the paper’s practical question. It is a legitimate modeling choice in isolation, but it is not a neutral one: it changes what parameter values can be studied at all.

3.2 Difference 2: cross-factor off-diagonal

Status: verified.

`orig` assigns `modelparam$c.cfct` as a constant to every cross-factor pair at distinct occasions (line 140). `covar` assigns `c.cfct * rho^tg`, decaying in the lag `tg`.

Arguments for the constant form. Again, fidelity to the reference. It also imposes no assumption about how quickly the coupling between distinct latent factors decays, which is a quantity for which little empirical guidance exists.

Arguments against. It is internally inconsistent with any decaying within-factor structure, since it asserts that a factor’s correlation with a different factor eight occasions away is as strong as with itself at the adjacent occasion. It contributes to the PD constraint in the same direction as Difference 1, since constant off-diagonal blocks raise the matrix’s dependence on its smallest eigenvalue.

Assessment. Secondary to Difference 1 in magnitude, and in the same direction. It is not separately identifiable from Difference 1 in any comparison run to date, since the two always vary together.

3.3 Difference 3: the interaction vector b

Status: verified, and the paper’s description is incorrect.

Appendix B.6 states that `orig` gates the interaction on whether the response mean is nonzero, giving a two-valued step of $\{0, c_{bm}\}$, and that at `t_half = 1.0` in the Hybrid design the vector becomes `c_bm * 1_n`, carrying no on-drug

versus off-drug contrast at all. The worked example at B.5 prints `b^orig = (0.4500, 0.4500, 0.4500, 0.4500)`.

That describes the following code, which is present in the vendored file but **commented out**, under the header `## Following commented out in Ron Thomas version:`

```
#if(means[which(n1==labels)]!=0){
#  correlations[n1,'bm']<-modelparam$c.bm
#  correlations['bm',n1]<-modelparam$c.bm
#}
```

The code that actually executes, under the header `## RON THOMAS VERSION:`, is line 153:

```
correlations["bm", n1] <- correlations[n1, "bm"] <-
  ifelse(brtest[p],
    ifelse(brmeans[p] == 0, 0, (mm1 / mm0) * modelparam$c.bm),
    modelparam$c.bm)
```

where `mm1` and `mm0` are the carryover-adjusted response means at occasions `p` and `p - 1`, and `brtest` is computed before the carryover adjustment, so it is `TRUE` exactly at off-drug occasions.

Re-running both code paths over the Hybrid path A design gives:

occasion	on drug	adj mu_BR	b active	b commented-out
OL1	yes	4.281	NA	0.45
OL2	yes	9.223	0.4500	0.45
BD1	yes	9.793	0.4500	0.45
BD2	yes	10.187	0.4500	0.45
BD3	no	2.547	0.1125	0.45
BD4	no	0.159	0.0281	0.45
COd	yes	4.281	0.4500	0.45
COp	no	0.017	0.0018	0.45

(`t_half = 1.0`, `scalefactor = 2`, `c_bm = 0.45`.)

The commented-out column reproduces the appendix exactly. The active column does the opposite: off-drug entries decay toward zero rather than rising to `c_bm`, and the on-drug versus off-drug contrast is preserved rather than lost.

The active decay belongs to the same exponential family as `covar`'s, but runs at twice the rate. The ratio `mm1 / mm0` inherits $(1/2)^{(\text{scalefactor} * t_{sd} / t_{half})}$ from the carryover adjustment at line 96, and with `scalefactor = 2` this is an effective half-life of `t_half / 2`, compounding multiplicatively across successive off-drug occasions. For comparison, at `t_half = 1.0` and one

week off drug, `covar` gives $0.45 * \exp(-\ln 2 * 1) = 0.225$ while `orig` gives 0.1125.

Arguments for the active form. It is graded rather than a step, which is arguably more plausible than a binary gate, and it ties the correlation channel to the same mean trajectory the carryover adjustment already modifies, which is a defensible internal consistency.

Arguments against. It is undocumented. Nothing in Paper 01, the vendored file's own header, or the directory README describes it. Its decay rate is set by `scalefactor`, a parameter whose own Roxygen comment in the vendored file reads `TODO update when understand what this does?`. And because it decays at twice `covar`'s rate, a comparison of the two at a matched nominal `t_half` is not a comparison at matched carryover.

Assessment. This is the most serious finding in this document. The paper's stated causal mechanism for `orig`'s behavior, that carryover opens its gate at off-drug occasions and destroys the contrast on which identification depends, is an accurate description of code that does not run. Any interpretation in Paper 01 resting on that mechanism requires revision, independent of anything else here.

3.4 Difference 4: the first occasion is never assigned

Status: verified.

The assignment loop at line 147 is guarded by `if (p > 1)`, and `correlations` is initialized as `diag(length(labels))` at line 106. The biomarker-response correlation at occasion 1 is therefore left at exactly zero under `orig`, regardless of `c_bm` and regardless of whether occasion 1 is on drug. In the table above this appears as the NA in the OL1 row. `covar` assigns all eight occasions.

Arguments for. None identified. This does not appear to be a modeling choice.

Arguments against. It removes one of eight occasions from the interaction channel, and it does so at an occasion that is on drug in every path of every design examined, so the loss falls on the most informative part of the series. It attenuates `orig`'s realized interaction relative to its nominal `c_bm` at every parameter setting. It also interacts with Difference 3: the `p > 1` guard exists because line 153 needs `p - 1` to form the ratio, so the defect is a direct consequence of the undocumented modification, not an independent bug.

Assessment. A defect rather than a difference in modeling philosophy. It is `c_bm`-independent and therefore is not addressed by restricting the correlation range.

4. The $c_{bm} \leq 0.35$ question

4.1 Why the question arises

Compound symmetry constrains the feasible correlation range more tightly than AR(1), as Appendix B.7 notes. If the binding objection to `orig` were only that it cannot represent large c_{bm} , then restricting attention to $c_{bm} \leq 0.35$ would neutralize the objection, and the choice between processes would come down to fidelity to the published reference, which would favor `orig`.

4.2 Positive-definiteness sweep

Status: verified. Minimum eigenvalue of Sigma, minimized across each design's randomization paths, at the parameter settings in Section 2.

Largest c_{bm} on the tested grid at which Sigma remains PD:

design	t_half	orig	covar
CO	0.0	0.30	0.60
CO	0.5	0.30	0.60
CO	1.0	0.30	0.60
Hybrid	0.0	0.30	0.45
Hybrid	0.5	0.35	0.50
Hybrid	1.0	0.35	0.50
OL+BDC	0.0	0.30	0.45
OL+BDC	0.5	0.35	0.45
OL+BDC	1.0	0.35	0.50

Minimum eigenvalue at exactly $c_{bm} = 0.35$:

design	t_half	orig	covar	orig PD
CO	0.0	-0.3069	8.516	no
CO	0.5	-0.3069	8.516	no
CO	1.0	-0.3069	8.516	no
Hybrid	0.0	-0.3140	2.158	no
Hybrid	0.5	0.0831	2.297	yes
Hybrid	1.0	1.2913	2.340	yes
OL+BDC	0.0	-0.3217	2.156	no
OL+BDC	0.5	0.0869	2.300	yes
OL+BDC	1.0	1.4385	2.344	yes

Three observations follow.

First, $c_{bm} = 0.35$ is **not** a safe ceiling for `orig`. Sigma is non-PD in four of the nine design-by-half-life cells: all three CO cells, and the zero-carryover cells of both other designs. The safe ceiling across the full grid is 0.30, not 0.35.

Second, where `orig` is PD at 0.35, it is marginal. Minimum eigenvalues of 0.083 and 0.087 at `t_half = 0.5` sit three orders of magnitude below `covar`'s at the same cells, and two orders below `orig`'s own values at `c_bm = 0.10`. A margin that thin is not robust to changes in `rho`, `c.cfct`, the number of occasions, or the response parameters, none of which were varied in this sweep.

Third, the CO design is uniformly the binding case under `orig`, and its minimum eigenvalue does not vary with `t_half` at all. This is consistent with CO's off-drug occasions being spaced 2.5 weeks or more apart, so that the `orig` decay, at twice `covar`'s rate, has driven the off-drug `b` entries to effectively zero at every half-life examined.

4.3 Silent repair

Status: verified.

Both implementations call `make.positive.definite(sigma, tol = 1e-3)` when the PD check fails, with no warning, no error, and no record in the returned object:

```
if (makePositiveDefinite) {
  if (!is.positive.definite(sigma)) {
    sigma <- make.positive.definite(sigma, tol = 1e-3)
  }
}
```

The consequence is that a non-PD cell does not fail. It silently becomes a different DGP. Measuring the realized biomarker-response correlations after repair, for CO path A at `t_half = 1.0`:

nominal c_bm	min eigenvalue	realized max	repaired
0.20	5.5645	0.2000	no
0.25	5.0361	0.2500	no
0.30	3.6264	0.3000	no
0.35	0.8645	0.3500	no
0.40	-2.8964	0.3955	yes
0.45	-7.3385	0.4370	yes

The distortion is modest in magnitude, roughly one to three points, but it is undisclosed and it is not constant across the grid, so it introduces a systematic and unreported difference between nominal and realized effect size precisely in the cells where `orig` is under the most strain. Power reported against a nominal `c_bm` in a repaired cell is not power at that `c_bm`.

4.4 Answer

Restricting `c_bm` to 0.35 or below does not remove the case for `covar`.

Five reasons remain, in descending order of force.

1. **The restriction is not tight enough.** At exactly 0.35, `orig` is non-PD in four of nine cells, including every CO cell. A restriction that actually cleared the PD objection would be `c_bm` $\leq$ 0.30, and even then only if no other parameter moves.
2. **The undocumented `b` modification is `c_bm`-independent.** `orig`'s off-drug interaction decays at twice `covar`'s rate for any correlation strength. The two processes are therefore not matched on carryover at a matched nominal `t_half`, so a comparison between them confounds carryover severity with everything else.
3. **The dropped first occasion is `c_bm`-independent.** It removes an on-drug occasion from the interaction channel at every parameter setting, including the null.
4. **Silent repair breaks the correspondence between nominal and realized effect size** in exactly the region where a restricted study would be operating closest to the boundary.
5. **The fidelity argument does not survive inspection.** The strongest reason to prefer `orig` at any `c_bm` is that it reproduces the published reference. But the vendored file is not the published reference. Its interaction assignment has been replaced, and the original is commented out beneath it. Whatever `orig` currently reproduces, it is not Hendrickson et al. as published.

Section 5 qualifies the first of these five reasons. A crossed decomposition shows the PD objection is narrower than it appears: it is an objection to compound symmetry alone, and one change fixes it. The other four reasons are not PD arguments and are unaffected.

The fifth point deserves emphasis because it inverts the usual trade-off. One would ordinarily accept a narrower feasible parameter range as the price of matching a published method. Here that price is being paid without the benefit being received.

4.5 What would change the answer

The case for `orig` would be materially stronger under any of the following, none of which currently holds.

- The vendored file is restored to the genuine upstream code, with the `RON THOMAS VERSION` block reverted and the original gate uncommented. `orig` would then serve its stated fidelity purpose, and Paper 01's Appendix B.6 would become an accurate description of it.

- The study is restricted to $c_{bm} \leq 0.30$ **and** to designs with $t_{half} > 0$ **and** the $p > 1$ guard is fixed. Under those three conditions the PD objection is genuinely resolved and the remaining differences are defensible modeling choices rather than defects.
- The correlation structure and the interaction gate are run as crossed factors, as Paper 01's own Appendix B.6 suggests. This would decompose the joint effect and permit a component-wise judgment rather than the present all-or-nothing comparison.

5. Decomposition: the minimum set of changes that fixes PD

5.1 Motivation and method

The stated reason for the **covar** changes was the non-positive-definite correlation matrix under **orig**. That justification implies a counterfactual which has not previously been tested: how many of the four changes are actually required to fix PD, and does each contribute in the direction assumed?

Status: verified. A single Sigma constructor was written with one flag per candidate change, reproducing **orig** when all four flags are off and **covar** when all four are on. The flags are:

- **ar1**: AR(1) within-factor ($\rho^{|w_i - w_j|}$) versus compound symmetry (constant ρ). Difference 1.
- **cfcd**: decaying cross-factor off-diagonal ($c_{cfct} * \rho^{|w_i - w_j|}$) versus constant c_{cfct} . Difference 2.
- **bexp**: exponential b ($c_{bm} * \exp(-\lambda * t_{sd})$) versus the ratio-graded **orig** form. Difference 3.
- **occ1**: assign the biomarker-response correlation at occasion 1 versus skipping it. Difference 4.

All 16 combinations were crossed with three designs and three half-lives. For each, c_{bm} was swept from 0.05 to 0.80 in steps of 0.01, and the largest value at which Sigma remained PD across every randomization path was recorded. The reported ceiling for a configuration is its worst cell over all nine design-by-half-life combinations.

5.2 Effect of each change in isolation

Starting from **orig** (all four flags off, worst-cell ceiling 0.34) and turning on exactly one flag:

change enabled	worst-cell ceiling	effect
none (orig)	0.34	baseline
ar1 only	0.48	+0.14
cfcd only	0.16	-0.18

change enabled	worst-cell ceiling	effect
bexp only	0.34	none
occ1 only	0.34	none

The result is unambiguous. **The AR(1) change alone accounts for the entire PD improvement.** The other three contribute nothing positive, and one of them is actively harmful.

5.3 The minimum set is a single change

`ar1` is both necessary and sufficient. Every configuration with `ar1` = TRUE reaches a worst-cell ceiling of 0.43 or above; every configuration with `ar1` = FALSE is capped at 0.34 or below, regardless of what the other three flags are set to. Ranked across all 16 configurations, the eight `ar1` = TRUE configurations occupy the top eight positions without exception.

Per design and half-life:

design	t_half	orig	AR(1) only	full covar
CO	0.0	0.34	0.59	0.61
CO	0.5	0.34	0.59	0.61
CO	1.0	0.34	0.59	0.61
Hybrid	0.0	0.34	0.49	0.45
Hybrid	0.5	0.35	0.51	0.50
Hybrid	1.0	0.36	0.54	0.54
OL+BDC	0.0	0.34	0.48	0.45
OL+BDC	0.5	0.35	0.49	0.49
OL+BDC	1.0	0.36	0.52	0.53

Changing only the within-factor correlation form, and leaving the other three orig behaviors untouched, yields a higher worst-cell ceiling (0.48) than the full set of four changes (0.45). The minimal fix is not merely sufficient; it is strictly better than `covar` on the criterion that motivated `covar` in the first place. AR(1)-only matches or exceeds full `covar` in seven of the nine cells, and trails it by at most 0.01 in the remaining two.

5.4 Why the cross-factor change is harmful

The `cfid` change, applied without `ar1`, lowers the ceiling from 0.34 to 0.16, the worst configuration tested. The mechanism is structural incoherence. Under compound symmetry a factor's correlation with itself is a constant `rho` at every lag. Introducing a decaying cross-factor term makes a factor's correlation with a *different* factor fall away with lag while its correlation with *itself* does not.

At long lags the matrix asserts strong within-factor coupling alongside near-zero cross-factor coupling, and the resulting block structure drives the smallest eigenvalue negative sooner than either uniform choice does.

Under `ar1` the same change is close to neutral, contributing at most 0.02 in either direction, because within-factor and cross-factor terms then decay together and the matrix stays internally consistent.

This is worth stating plainly: **Difference 2 was not a PD fix. In isolation it is a PD regression, and its only defensible justification is internal consistency with Difference 1.**

5.5 Why the `b` and `occ1` changes are irrelevant to PD

Neither `bexp` nor `occ1` moved the ceiling by a single grid step in any configuration. This is expected on inspection. The biomarker-response entries occupy one row and one column of a matrix with $2 + 3n$ dimensions, and their magnitudes are bounded by `c_bm` itself. The binding constraint comes from the $3n \times 3n$ within- and cross-factor blocks, not from the biomarker margin.

The practical consequence is that Differences 3 and 4 cannot be defended on PD grounds at all. Difference 3 is an undocumented change of decay rate (Section 3.3) and Difference 4 is a defect (Section 3.4). Whatever their merits, the non-positive-definiteness of the `orig` matrix is not among them.

5.6 Implications

1. **The minimum set of changes that fixes the PD problem is one change: compound symmetry to AR(1).** Adopting it alone recovers more feasible parameter range than the full four-change set.
2. **Three of the four changes in `covar` are not PD fixes.** They may be justified on other grounds, but the PD rationale does not extend to them and should not be offered for them.
3. **A minimal-change variant is worth constructing.** An `orig + AR(1)` process would isolate the correlation structure as the sole difference from the published reference, which is exactly the crossed-factor decomposition Paper 01's Appendix B.6 says would be required, and it is a one-line change to the vendored file.
4. **This changes the practical answer of Section 4.4 in one respect.** The PD objection to `orig` is narrower than it appeared: it is an objection to compound symmetry specifically, not to the `orig` process as a whole. The other four objections in Section 4.4 stand unchanged, since none of them is a PD argument.

6. Recommendations

1. **Do not adopt orig as a comparison arm on the strength of a restricted c_bm range.** The restriction addresses one of four differences and does not fully address even that one.
2. **Correct Paper 01, Appendix B.5 to B.7.** The **b** row, the worked example, and the third interpretive paragraph after the B.6 table describe code that does not run. This is independent of any decision about future simulation work and should be treated as a correctness fix.
3. **Verify the vendored files against the upstream repository** (github.com/rchendrickson/pmsimstats, commit 3035581). The directory README states the files are verbatim apart from one `data.table` indexing patch and that no other lines were changed. At least two further edits are visible in the file: the `RON THOMAS VERSION` block, and a typo fix at line 134 annotated `##### <-----FIXING TYPO, this was [n1,n2] again.` Either the README's provenance claim or the file needs correcting.
4. **Make PD repair loud.** Both implementations should record the repair in the returned object and warn. A silent change to the DGP is not acceptable in a simulation study whose outputs are reported against nominal parameter values.
5. **Fix the $p > 1$ guard** if `orig` is retained in any form, or document it as a known property of the arm if the intent is to reproduce whatever the vendored file does.
6. **Construct an orig + AR(1) variant.** Section 5 shows this is a one-line change that fixes PD more effectively than the full `covar` set. It isolates the correlation structure as the sole difference from the reference implementation and would let Paper 01 report a decomposition rather than a confounded bound.
7. **Stop offering the PD rationale for Differences 2, 3, and 4.** Section 5 shows none of them improves positive definiteness, and Difference 2 degrades it when applied on its own. If those changes are retained they need a different justification.
8. **If a full decomposition of the power results is wanted, run the crossed design.** Correlation structure (CS versus AR(1)) crossed with interaction gate (step versus graded) at `c_bm <= 0.30`, three designs, three half-lives. This is four DGP variants rather than two and would separate the effects the present comparison confounds. The PD half of that decomposition is already done (Section 5); the power half is not.

7. Limitations

The PD sweep used a single parameter set (`rho = 0.7`, `c.cf1t = 0.2`, `c.cfct = 0.1`, `N = 35`, eight occasions, the package's `extracted_rp` and `extracted_bp`). The PD boundary depends on all of these, and the boundaries reported in Section 4.2 should not be read as general. In particular, larger `rho` will tighten the `orig` boundary further.

The Section 5 decomposition is a statement about positive definiteness only. It establishes which changes widen the feasible `c_bm` range, not which produce better statistical behavior within that range. An `orig + AR(1)` process has a higher PD ceiling than `covar` but was not evaluated for power, Type I error, or estimator bias, and nothing here implies it would perform better on those.

The Section 5 constructor reproduces both processes from the source read in Section 2 rather than calling them. It was checked against the `orig` and `covar` ceilings obtained in Section 4.2 from the real implementations, which agree to within the coarser grid step used there, but it is a reimplement and may diverge in behavior not exercised by this sweep.

No power simulation was run for this document. The consequences of Differences 3 and 4 for realized power under `orig` are argued from the structure of the covariance matrix, not measured. Quantifying them would require running the `orig` arm with and without the `p > 1` guard, and with the commented-out gate restored, which is the natural follow-up.

The upstream provenance question is unresolved. The comparison against github.com/rchendrickson/pmsimstats was not performed, and the attribution of the RON THOMAS VERSION edits is therefore inferred from the annotation text alone.

`implementations/original/R/generateData.R`, the package's other copy of the reference implementation, was inspected and found to differ from the vendored copy: it gates on `d[p]$onDrug` and uses `c.cfct * rho^tg` for the cross-factor off-diagonal. It is therefore a third variant, distinct from both `covar` and the vendored `orig`, and its relationship to the other two was not pursued here.

8. Provenance

Audit conducted 2026-09-08 against the working tree at commit `3ce1162`. Computations executed on the host R session (R 4.6.1), outside the project container. Scratch scripts were not retained in the repository.